Single Instruction Multiple Data – ARM NEON

Sorbonne Université – Master SESI – MU4IN112 – PPSE

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Single Instruction Multiple Data (SIMD)

Example of scalar instructions:

```
loop:
  load a, [p_a]
  load b, [p_b]
  add c, a, b # c[i] = a[i] + b[i]
  store [p_c], c
  add p_a, p_a, 4 # 4 bytes, 32-bit
  add p_b, p_b, 4 # 4 bytes, 32-bit
  add p_c, p_c, 4 # 4 bytes, 32-bit
  add i, i, 1
jumple i, 10000, loop
```

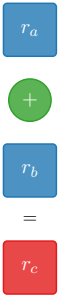
Example of SIMD instructions:

```
loop:
  vload va, [p_a] # read 4 elements
  vload vb, [p_b] # read 4 elements
  vadd vc, va, vb # add 4 elements
  vstore [p_c], vc # write 4 elements
  add p_a, p_a, 4*4 # 4*4 bytes, 128-bit
  add p_b, p_b, 4*4 # 4*4 bytes, 128-bit
  add p_c, p_c, 4*4 # 4*4 bytes, 128-bit
  add i, i, 4 # increment i of 4
jumple i, 10000, loop
```

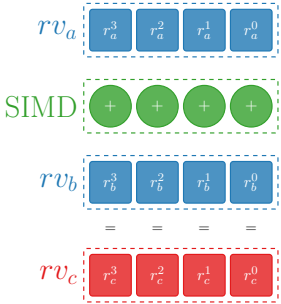
- The `vadd`, `vload` and `vstore` instructions perform operations on 4 elements at a time
  - 4 times fewer instructions (potential acceleration of 4)
  - The number of elements in a vector register = the # lanes (here 4)
  - Each element of `vc[i..i+4]` is computed in parallel

Single Instruction Multiple Data (SIMD)

- Scalar instruction: produces data in 1 cycle (to simplify)



- A SIMD instruction produces n data in 1 cycle (to simplify)



- SIMD instructions operate on so-called "vector" registers (rv_a, rv_b, rv_c)

SIMD and Vector Instruction Sets – Part 1

Date	ISA	Type	Micro-architecture	Registers Size
1997	MMX	SIMD	Intel Pentium	64-bit
1999	Altivec	SIMD	Apple+IBM PowerPC	128-bit
1999	SSE	SIMD	Intel Pentium III	128-bit
2000	SSE2	SIMD	Intel Pentium IV	128-bit
2004	SSE3	SIMD	Intel Pentium IV (Prescot)	128-bit
2005	NEONv1	SIMD	ARMv7	128-bit
2008	SSE4	SIMD	Intel Core 2 Duo	128-bit
2011	AVX	SIMD	Intel Sandy Bridge (Core i)	256-bit
2012	NEONv2	SIMD	ARMv8	128-bit
2013	AVX2	SIMD	Intel Haswell (Core iX)	256-bit
2016	AVX-512	SIMD	Intel Xeon Phi	512-bit
2017	SVE	Vector	ARMv8	–
2019	SVE2	Vector	ARMv9	–
2021	RVV	Vector	RISC-V	–

SIMD and Vector Instruction Sets – Part 2

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2021	RVV	Vector	RISC-V	–

- A large number of existing instruction sets
- Some are backward-compatible
  - AVX-512 → AVX → SSE → MMX
  - NEONv2 → NEONv1
- Most instruction sets are incompatible with each other
  - Each instruction set is written differently

Assembly Code

Example of single precision floating-point addition according to different ISAs:

x86 SSE

```
addps xmm, xmm
```

ARM NEON

```
FADD Vd.4S,Vn.4S,Vm.4S
```

x86 AVX

```
vaddps ymm, ymm, ymm
```

ARM SVE

```
FADDP Zresult.S, Pg.M, Zresult.S, Zop2.S
```

x86 AVX-512

```
vaddps zmm, k, zmm, zmm
```

RISC-V Vector extension

```
vadd.vv vd, vs2, vs1, vm
```

- As many different ways as there are instruction sets
- Assembly code is not portable :-(
- A major productivity brake

Intrinsic Functions: Example of Addition

x86 AVX

```
__m256 _mm256_add_ps (__m256 a, __m256 b);
```

x86 AVX-512

```
__m512 _mm512_mask_add_ps (__m512 src, __mmask16 k, __m512 a, __m512 b);
```

ARM NEON

```
float32x4_t vaddq_f32 (float32x4_t a, float32x4_t b);
```

ARM SVE

```
svfloat32_t svaddp_f32_x (svbool_t pg, svfloat32_t op1, svfloat32_t op2);
```

RISC-V Vector extension

```
vfloat32m1_t vfadd_vv_f32m1_m (vbool32_t mask, vfloat32m1_t maskedoff, vfloat32m1_t op1, vfloat32m1_t op2, size_t vl);
```

Intrinsic Functions

Definition: An intrinsic function is a function to which one (and only one) assembly instruction corresponds.

- As many different ways as there are instruction sets
- Code with intrinsic functions is not portable
- Better productivity than pure assembly programming
  - The compiler takes care of register allocation
  - Easy to interface with C and C++ code
  - Often the chosen method by embedded system developers

Work with the Compiler

- In some cases, the compiler can automatically vectorize the source code
  - Works when computations are regular and with `-O3` optimization level (GCC)
  - You can display a vectorization report to see which parts of the code have actually been vectorized
- The compiler often does not vectorize well
  - Either it does not vectorize at all
  - Or it vectorizes a little, but we could do much better
- It is possible to help the compiler with annotations (ex. OpenMP SIMD)
- In HPC, a lot of code uses the compiler + annotations
  - Supercomputers evolve fast
  - Relying on the compiler guarantees source code portability

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Source of Inspiration

This section of is strongly inspired by the excellent slides of Pr. Lionel Lacassagne (Sorbonne University).

Special thanks to him for sharing its work!

Strongly-typed Data Types

- C89 and before
  - { `signed`, `unsigned` } × { `char` }, { `short`, `long` } × { `int` }, `float`, `double`
  - ➔ `unsigned short`, `signed long int`, ...
  - Major portability problem: type size can change depending on arch. and OS
- From C99 (`stdint.h`)
  - Fixed-width integer types: `int8_t`, `int16_t`, `int32_t`, `int64_t`, `uint8_t`, `uint16_t`, `uint32_t`, `uint64_t`
  - But the fixed-width floating-point data types are missing
- ARM choose to use the same philosophy
  - `int8_t`, `int16_t`, `int32_t`, `int64_t`, `uint8_t`, `uint16_t`, `uint32_t`, `uint64_t`, `float16_t`, `float32_t`, `float64_t`
  - ➔ Fixed-width floating-point data types are present!

Strongly-typed SIMD Data Types – Part 1

SIMD types use strong typing: `<type><size>x<number of lanes>_t`

- 64-bit registers
  - Signed integers: `int8x8_t`, `int16x4_t`, `int32x2_t`, `int64_t`,
  - Unsigned integers: `uint8x8_t`, `uint16x4_t`, `uint32x2_t`, `uint64_t`,
  - Floating-point numbers: `float16x4_t`, `float32x2_t`, `float64_t`
  - Polynomial: `poly8x8_t`, `poly16x4_t`
- 128-bit registers
  - Signed integers: `int8x16_t`, `int16x8_t`, `int32x4_t`, `int64x2_t`,
  - Unsigned integers: `uint8x16_t`, `uint16x8_t`, `uint32x4_t`, `uint64x2_t`,
  - Floating-point numbers: `float16x8_t`, `float32x4_t`, `float64x2_t`
  - Polynomial: `poly8x16_t`, `poly16x8_t`

Strongly-typed SIMD Data Types – Part 2

SIMD types use strong typing: `<type><size>x<number of lanes>_t`

The availability of the different types and their associated instructions depends on the architecture version ({ v7, v8 } × { AArch32, AArch64 }), their specialization (ARM v8.2 for `float16_t`) and the implementations by designers (Apple, Nvidia, Samsung, Qualcomm, Huawei, ...).

Type of Instructions

4 main types of SIMD instructions:

1. Initialization and memory access
   - `load`, `store`, `set`
2. Arithmetic and logic
   - `add`, `mul`, `div`, `shift`, ...
   - `and`, `or`, `xor`, `not`
3. Comparison
   - $<$, $\leq$, $=$, $\neq$, $\geq$, $>$ (Fortran mnemonics: lt, le, eq, neq, ge, gt)
4. Shuffle & permutation
   - to reorganize values in a SIMD register, or in memory
   - because without it, some computations would be inefficient in SIMD

Type of Instructions

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It is not required to know all the intrinsics but we need to be able to find the instruction we are looking for from the technical documentation.

Initialization – vdup_n / vmov_n

- Set of values (or variables)
  - C++ style: `T x = {...};`
  - Example: `uint32x4_t x = {1, 2, 3, 4};` ➔ `x = [1, 2, 3, 4]`
  - Possible with scalar variables of compatible type (in this case `uint32_t`)
- Unique value (or unique variable) – `vdup_n_T`, `vdupq_n_T`
  - `uint32_t u = 1;`
  - Example 64-bit: `uint32x2_t x = vdup_n_u32(1);` ➔ `x = [1, 1]`
  - Example 128-bit: `uint32x4_t x = vdupq_n_u32(u);` ➔ `x = [1, 1, 1, 1]`
- Unique value (or unique variable) – `vmov_n_T`, `vmovq_n_T`
  - `uint32_t u = 1;`
  - Example 64-bit: `uint32x2_t x = vmov_n_u32(1);` ➔ `x = [1, 1]`
  - Example 128-bit: `uint32x4_t x = vmovq_n_u32(u);` ➔ `x = [1, 1, 1, 1]`

Access to a Lane – vset_lane / vget_lane

- Set – `dst = vset_lane_T(val, src, i)`
  - Copy `src` into `dst` and write `val` in the `i`-lane of `dst`
  - To modify a register within a loop (efficiently)
- Get – `val = vget_lane_T(src, i)`
  - Extract the `i`-element
- Example

```
uint32x4_t x = (uint32x4_t) {10, 11, 12, 13}; // x = [10 11 12 13]
uint32x4_t y = vsetq_lane_u32(14, x, 3);      // y = [10 11 12 14]
uint32_t   t = vgetq_lane_u32(y, 2);          // t = 12
```

Memory Access – vld1 / vst1

- LoaD – `vld1_T(addr)`, `vld1q_T(addr)`
- STore – `vst1_T(addr, x)`, `vst1q_T(addr, x)`

```
// 64-bit loads ('T8', 'T16' and 'T32' are data pointers, 'r<T>_64' are 64-bit registers)
uint8x8_t  r8_64  = vld1_u8 (T8 ); // [0, 1, 2, 3, 4, 5, 6, 7]M
uint16x4_t r16_64 = vld1_u16(T16); // [01, 23, 45, 67]M
uint32x2_t r32_64 = vld1_u32(T32); // [0123, 4567]M
// --------------------------
// 128-bit stores ('T8', 'T16' and 'T32' are data pointers, 'r<T>_128' are 128-bit registers)
vst1q_u8 (T8 , r8_128 ); // [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F]M
vst1q_u16(T16, r16_128); // [01, 23, 45, 67, 89, AB, CD, EF]M
vst1q_u32(T32, r32_128); // [01234567, 89ABCDEF]M
```

- The basic memory access instructions manipulate 1 set of elements, but there are also instructions that manipulate 2, 3 or even 4 sets of elements
  - For strided accesses 1:2, 1:3, 1:4
  - For conversions: Array-of-Struct - Struct-of-Array (AoS / SoA)

Display for Debug

- No working code without debugging!
- Here is an example of printing SIMD register values
  - 8-bit elements & 128-bit SIMD register ➔ 16 elements

```
1 void display_r8_128(uint8x16_t r8_128, char *format) {
2     uint8_t T[16];                // declare and allocate array in memory
3     vst1q_u8(T, r8_128);          // write the vector register in memory
4     for (int i = 0; i < 16; i++) // for each element of the SIMD register
5       printf(format, T[i]);       // print the value of lane i
6 }
```

- SIMD register elements (= lanes) displayed in ascending order
  - As if the data were in memory
  - Same display on little and big endian architectures

Binary Operations – `not` / `or` / `and` / `xor`

- Basic operators
  - NOT: `vmovn` (MOVe Not)
  - AND: `vand`
  - OR: `vorr` (OR Register)
  - XOR: `veor` (Exclusive OR)
- Advanced / specialized operators
  - OR NOT: `vorn` (OR Not)
  - AND NOT: `vbic` (BIt Clear)
  - REV: `vrbit` (Reverse BIT) inverts the bits of each element
  - XOR[2]: `veor3` *3-way exclusive OR*
  - `vbcax` *bit clear and exclusive OR*

Arithmetic Operations – `add` / `sub` / `mul` / `div`

- Classic arithmetic operations
  - ADD: `vadd` 8, 16, 32, 64
  - SUB: `vsub` 8, 16, 32, 64
  - NEG: `vneg` 8, 16, 32, 64
  - MUL: `vmul` 8, 16, 32, 64
  - DIV: `vdiv` 16, 32, 64 (only for floats)
- Specialized arithmetic operations
  - MUL: `vmul_lane(a, b, i)`: $a \times b_i$
  - MUL: `vmul_n(v, s)`: multiplication by a scalar value

Saturated Arithmetic Operations – Integers

1. General case of arithmetic: Size change (C and mathematics)
   - Addition $n + n \rightarrow n + 1$ bits so $2n$ bits
   - Multiplication: $n \times n \rightarrow 2n$ bits
   - $\Rightarrow$ Loss of parallelism ($\div 2$)
   - The size change prevents the compiler from vectorizing integer codes
   - Example `(a + b + c + d) / 4`
2. Special case #1: Without size change
   - Addition: $n + n \rightarrow n$ bits because info on missing addition holdback
   - Multiplication: $n \times n \rightarrow 2n$ because operands on n bits
   - Signal and image processing
   - Bad example `(a / 4 + b / 4 + c / 4 + d / 4)` (loss of precision)
3. Special case #2: With only a few holdbacks
   - Often the case with audio
   - Computations without loss of parallelism + saturation of results that exceed (internal holdback test)

Saturated Arithmetic Ops – vqadd / vqsub

- Only for addition (`vqadd`) and subtraction (`vqsub`)
  - Unsigned saturated arithmetic `vqadd_u`, `vqsub_u` $[0, 2^n - 1]$
  - Signed saturated arithmetic `vqadd_s`, `vqsub_s` $[-2^{n-1}, 2^{n-1} - 1]$

```
uint8x8_t u = (uint8x8_t) {1, 2, 3, 4, 5, 6, 7, 8};
                          // [  1   2   3   4   5   6   7   8]
u = vqadd_u8(u, u); // [  2   4   6   8  10  12  14  16]
u = vqadd_u8(u, u); // [  4   8  12  16  20  24  28  32]
u = vqadd_u8(u, u); // [  8  16  24  32  40  48  56  64]
u = vqadd_u8(u, u); // [ 16  32  48  64  80  96 112 128]
u = vqadd_u8(u, u); // [ 32  64  96 128 160 192 224 255] <- sat(128 + 128) = sat(256) = 255
u = vqadd_u8(u, u); // [ 64 128 192 255 255 255 255 255]
u = vqadd_u8(u, u); // [128 255 255 255 255 255 255 255]
u = vqadd_u8(u, u); // [255 255 255 255 255 255 255 255]
```

Ordering instructions – `min` / `max` – Part 1

- Two missing scalar instructions
  - In scalar mode, use a control structure: `if(a<b) m=a; else m=b`
  - The extremum is computed by subtraction and comparison of the sign of the result: `s=a-b; if(s<0) m=a; else m=b;`
  - On simple processors, the control structure generates a pipeline stalls
  - But on complex processors, there is a conditional `move` (= `cmove`)
- In SIMD
  - The extremum is computed without any control structure
  - SIMD version more is efficient

Ordering instructions – `min` / `max` – Part 2

Extraction of min and max scalars from 2 arrays

- Array version (v1) (most compact)

```
for (int i = 0; i < n; i++) {
  if (A[i] < B[i]) {
    Min[i] = A[i]; Max[i] = B[i];
  } else {
    Min[i] = B[i]; Max[i] = A[i];
  }
}
```

- Version with ternary conditions (v2)

```
for (int i = 0; i < n; i++) {
  uint8_t a = A[i];
  uint8_t b = B[i];
  uint8_t m = (a < b) ? a : b; // better without if
  uint8_t M = (a < b) ? b : a; // 'cmov' instr.
  Min[i] = m; Max[i] = M;
}
```

- SIMD version

```
for (int i = 0; i < n; i += 16) {
  uint8x16_t a = vld1q_u8(A + i);
  uint8x16_t b = vld1q_u8(&B[i]);
  uint8x16_t m = vminq_u8(a, b);
  uint8x16_t M = vmaxq_u8(a, b);
  vst1q_u8(Min + i, m);
  vst1q_u8(&Max[i], M);
}
```

— SIMD code is easy to write from the scalar version 2

Comparison instructions – `vc`

- SIMD intrinsic names are derived from Fortran mnemonics
  - For integers and floats
  - `vclt <` LT: Less Than
  - `vcle <=` LE: Less or Equal
  - `vceq ==` EQ: EQual
  - `vcge >=` GE: Greater or Equal
  - `vcgt >` GT: Greater Than
- Extension to unary (monoadic) operators
  - Avoids the use of a second register initialized to zero
  - For integers signed and floats
  - `vcltz` < 0
  - `vclez` ≤ 0
  - `vceqz` $= 0$
  - `vcgez` ≥ 0
  - `vcgtz` > 0

Web References

- Neon Intrinsics Reference
  https://developer.arm.com/architectures/instruction-sets/simd-isas/neon/intrinsics
- Armv8 Instruction Set Architecture: https://developer.arm.com/docs/ddi0596/c/simd-and-floating-point-instructions-alphabetic-order
- Coding for NEON
  - Part 1: Load and Stores https://community.arm.com/developer/ip-products/processors/b/processors-ip-blog/posts/coding-for-neon---part-1-load-and-stores
  - Part 2: Dealing With Leftovers https://community.arm.com/developer/ip-products/processors/b/processors-ip-blog/posts/coding-for-neon---part-2-dealing-with-leftovers
  - Part 3: Matrix Multiplication https://community.arm.com/developer/ip-products/processors/b/processors-ip-blog/posts/coding-for-neon---part-3-matrix-multiplication
  - Part 4: Shifting Left and Right https://community.arm.com/developer/ip-products/processors/b/processors-ip-blog/posts/coding-for-neon---part-4-shifting-left-and-right
  - Part 5: Rearranging Vectors https://community.arm.com/developer/ip-products/processors/b/processors-ip-blog/posts/coding-for-neon---part-5-rearranging-vectors

Q&A

Thank you for listening!
Do you have any questions?